

# Lecture – 3

## CS 372 (Computer Graphics)



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# NUMERICAL

Draw a line from (5,8) to (9,11) using Bresenham algorithm.

$$dy = 11 - 8 = 3; dx = 9 - 5 = 4; d = 2dy - dx = 2 \times 3 - 4 = 2$$

$$(\Delta d)_E = 2dy = 2 \times 3 = 6; (\Delta d)_{NE} = 2(dy - dx) = 2(3 - 4) = -2.$$

$$S1: d = 2 > 0, NE, x' = x + 1, y' = y + 1, x' = 6, y' = 9.$$

$$d = d + (\Delta d)_{NE} = 2 - 2 = 0$$

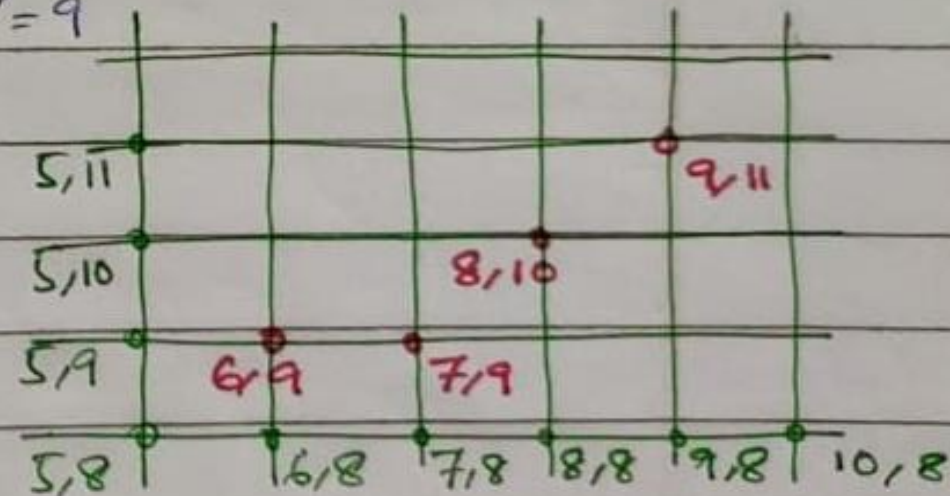
$$S2: d = 0, E, x' = x + 1, y' = y, x' = 7, y' = 9$$

$$d = d + (\Delta d)_E = 0 + 6 = 6$$

$$S3: d = 6 > 0, NE, x' = 8, y' = 10$$

$$d = d + (\Delta d)_{NE} = 6 - 2 = 4$$

$$S4: d = 4 > 0, NE, x' = 9, y' = 11.$$



# OCTANTS

12

Octant

Change

1

None

2

Switch  $x$  &  $y$

(Line having slope  $> 1$ . So more movement along  $y$ )

3

Switch  $x$  &  $y$  and 4

(Slope  $> 1$  and -ve. Also drawing right to left)

4.

Draw right to left, use 8.

5.

Draw right to left,

6.

Draw right to left, use 2

7.

Switch  $x$  &  $y$ , use 8

8.

Use  $y = y - 1$ .

